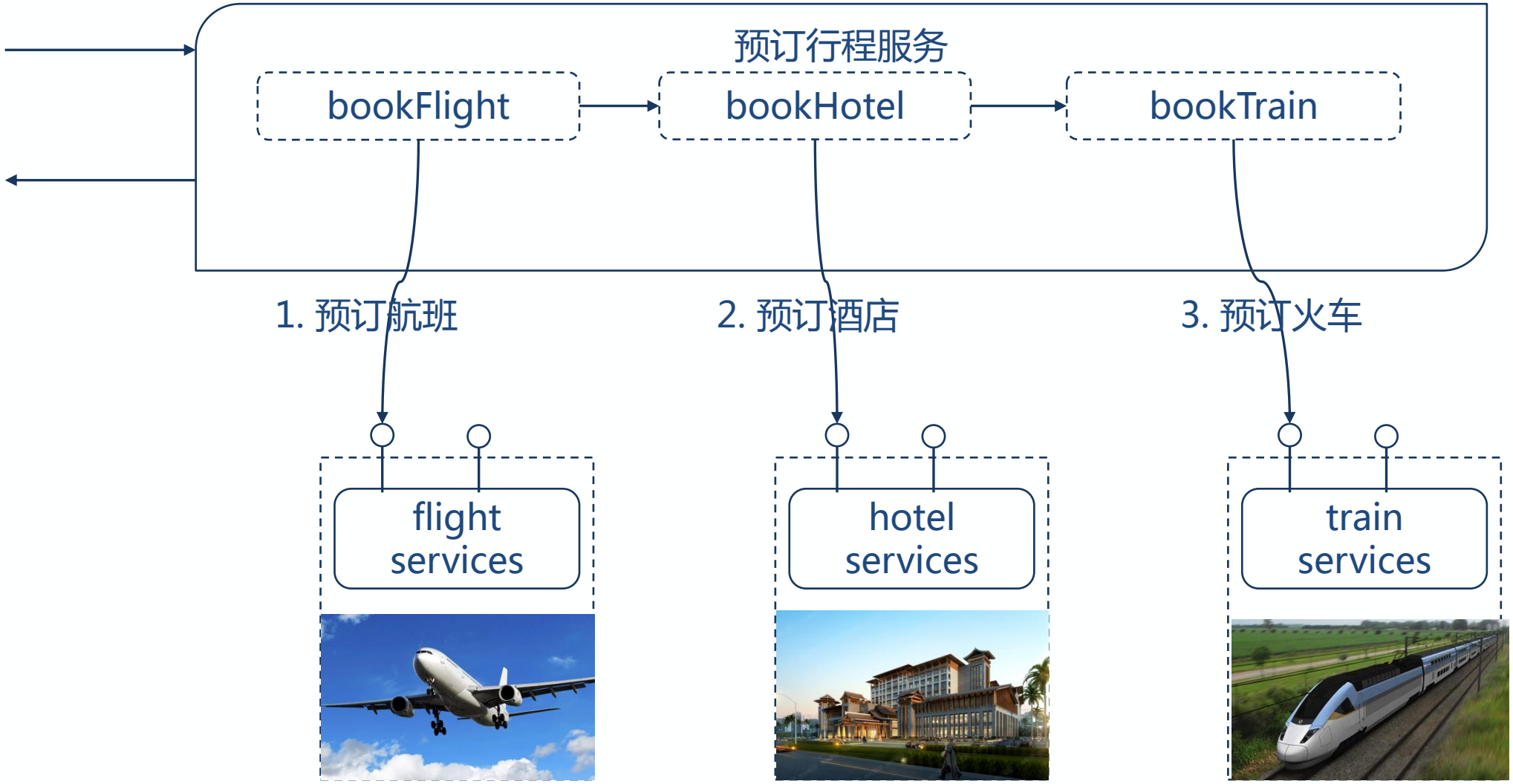
微服务架构下如何保证数据一致性（三）

田向阳

内容简介

- 实现补偿模式的关键在于业务流水的记录
- 通过重试保证补偿过程的完整，从而满足最终一致性
- TCC模式是优化的补偿模式

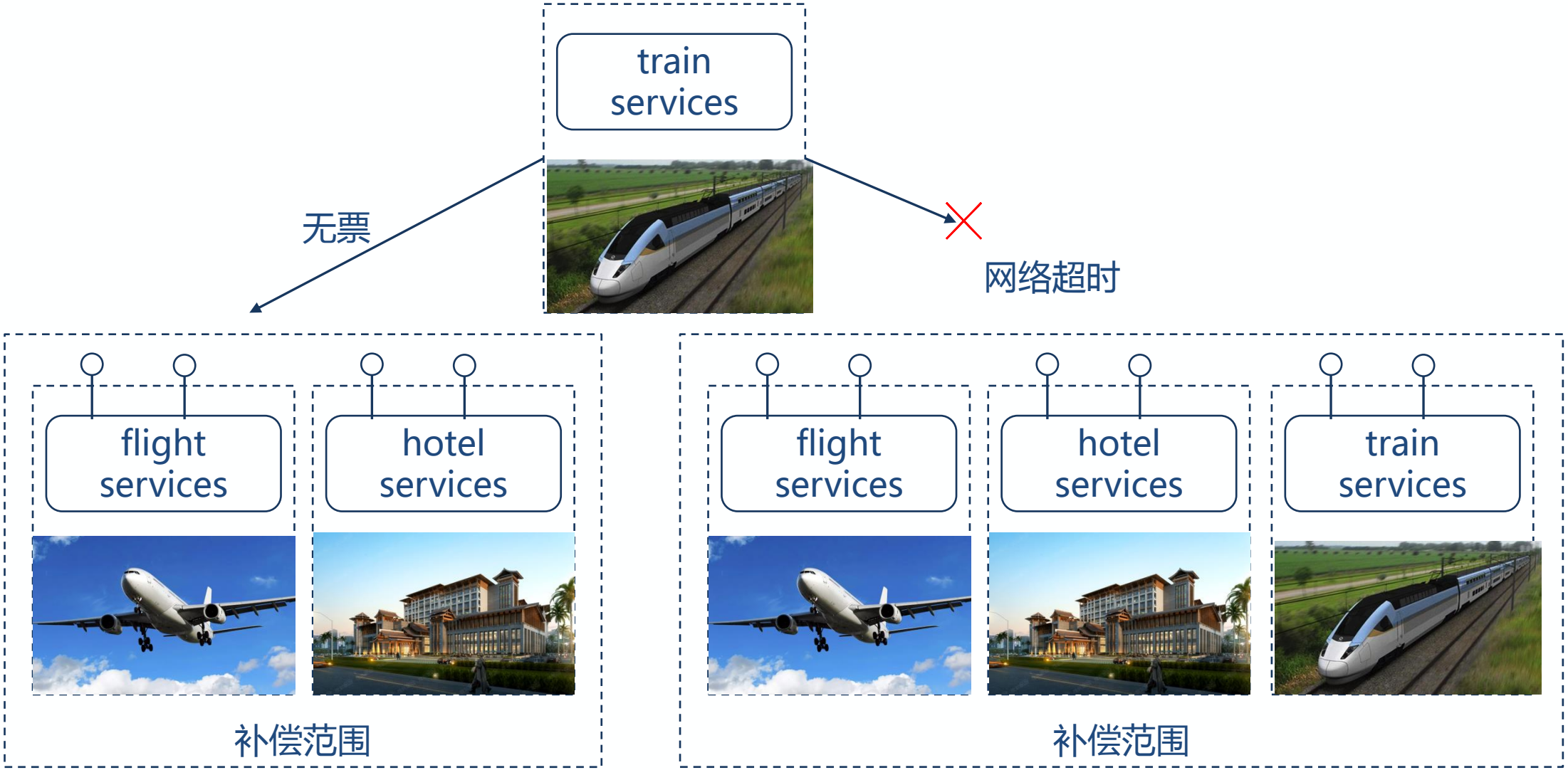
补偿模式


预订行程服务
bookFlight
bookHotel
bookTrain
1. 预订航班
2. 预订酒店
3. 预订火车
flight services
hotel services
train services

补偿模式

预订行程服务
bookFlight
bookHotel
bookTrain
bookHotel
bookHotel
1. 预订航班
2. 预订酒店
3. 预订火车
5. 取消航班预订
4. 取消酒店预订
flight services
hotel services
train services

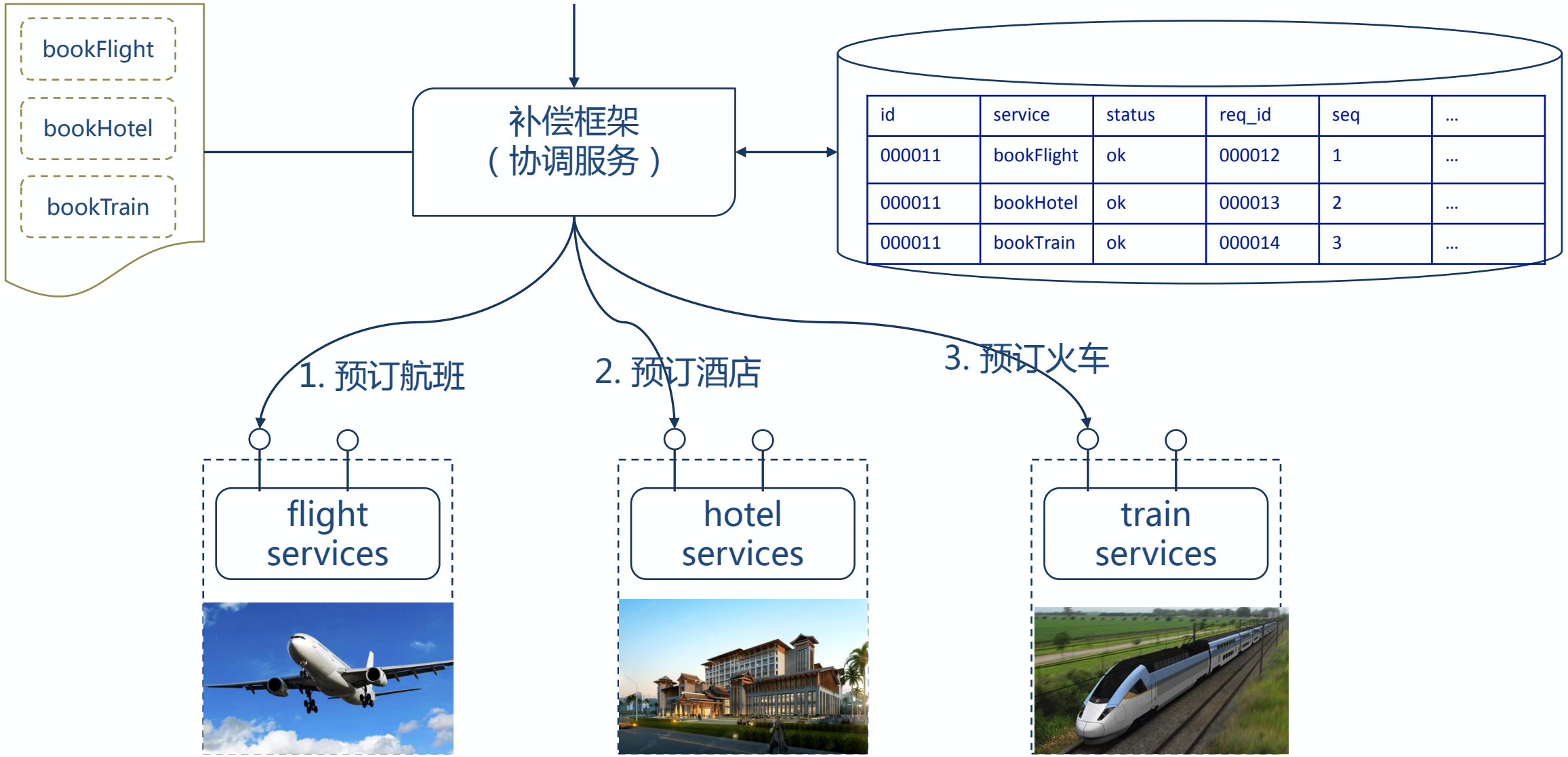
补偿范围跟异常原因有关



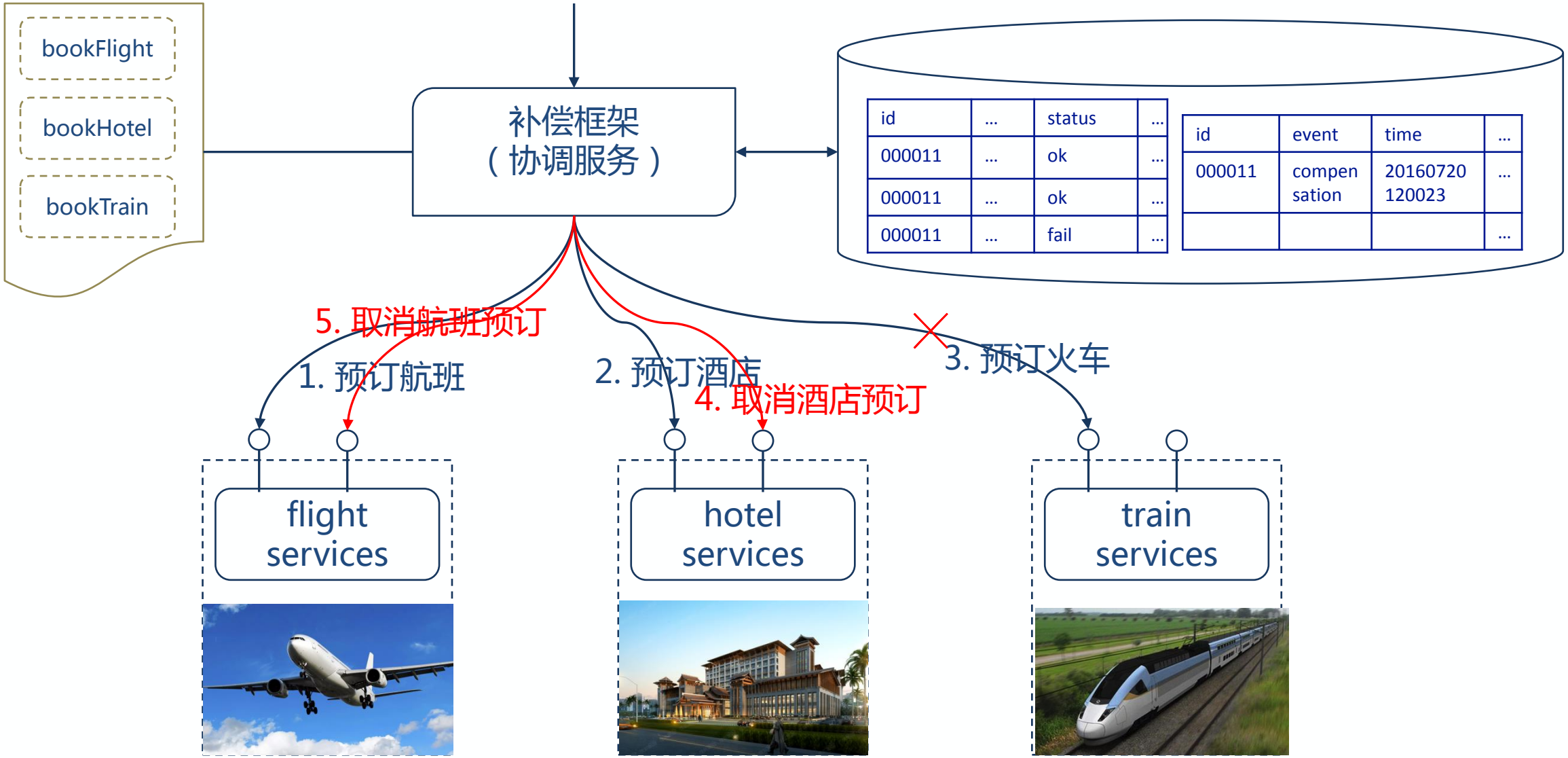
补偿模式

实现补偿模式的关键在于业务流水的记录

补偿模式


bookFlight
bookHotel
bookTrain
补偿框架
（协调服务）
id | service | status | req_id | seq | ...
000011 | bookFlight | ok | 000012 | 1 | ...
000011 | bookHotel | ok | 000013 | 2 | ...
000011 | bookTrain | ok | 000014 | 3 | ...
1. 预订航班
2. 预订酒店
3. 预订火车
flight services
hotel services
train services

补偿模式


bookFlight
bookHotel
bookTrain
补偿框架
（协调服务）
id | ... | status | ...
000011 | ... | ok | ...
000011 | ... | ok | ...
000011 | ... | fail | ...
id | event | time | ...
000011 | compensation | 20160720120023 | ...
5. 取消航班预订
1. 预订航班
2. 预订酒店
4. 取消酒店预订
3. 预订火车
flight services
hotel services
train services

大表VS关联表

trace

ID	service	seq	status	col1	col2	col3	…
1	bookFlight	1	send	2	10000	xxxx	

大表

关联表

platform_trace

ID	…	b_tbname	b_keys	b_keys_v
1		bank_trace	id	2

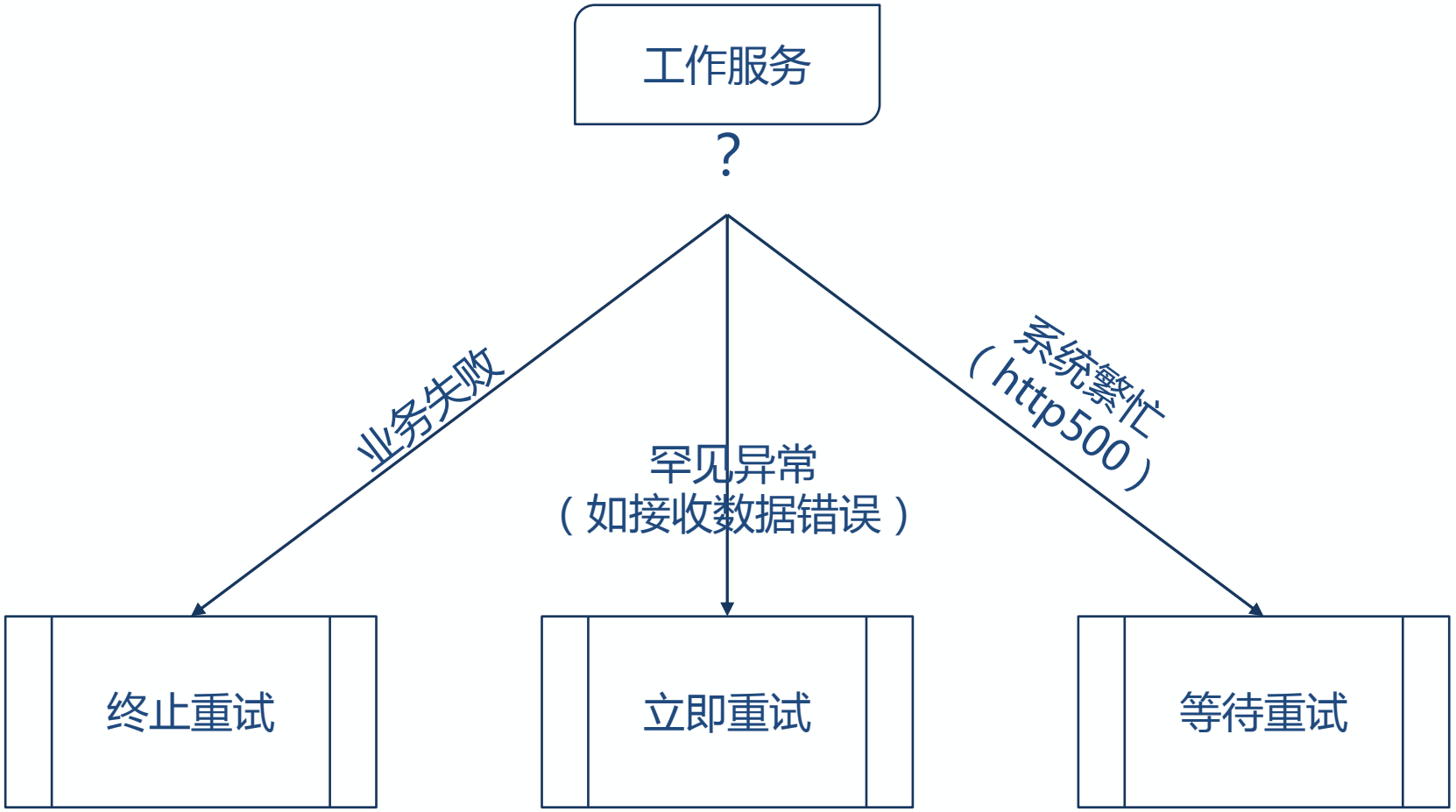
bank_trace

id	amount	card	time	…
2	10000	xxxx	xxxx	

补偿模式

通过重试保证补偿过程的完整，从而满足最终一致性

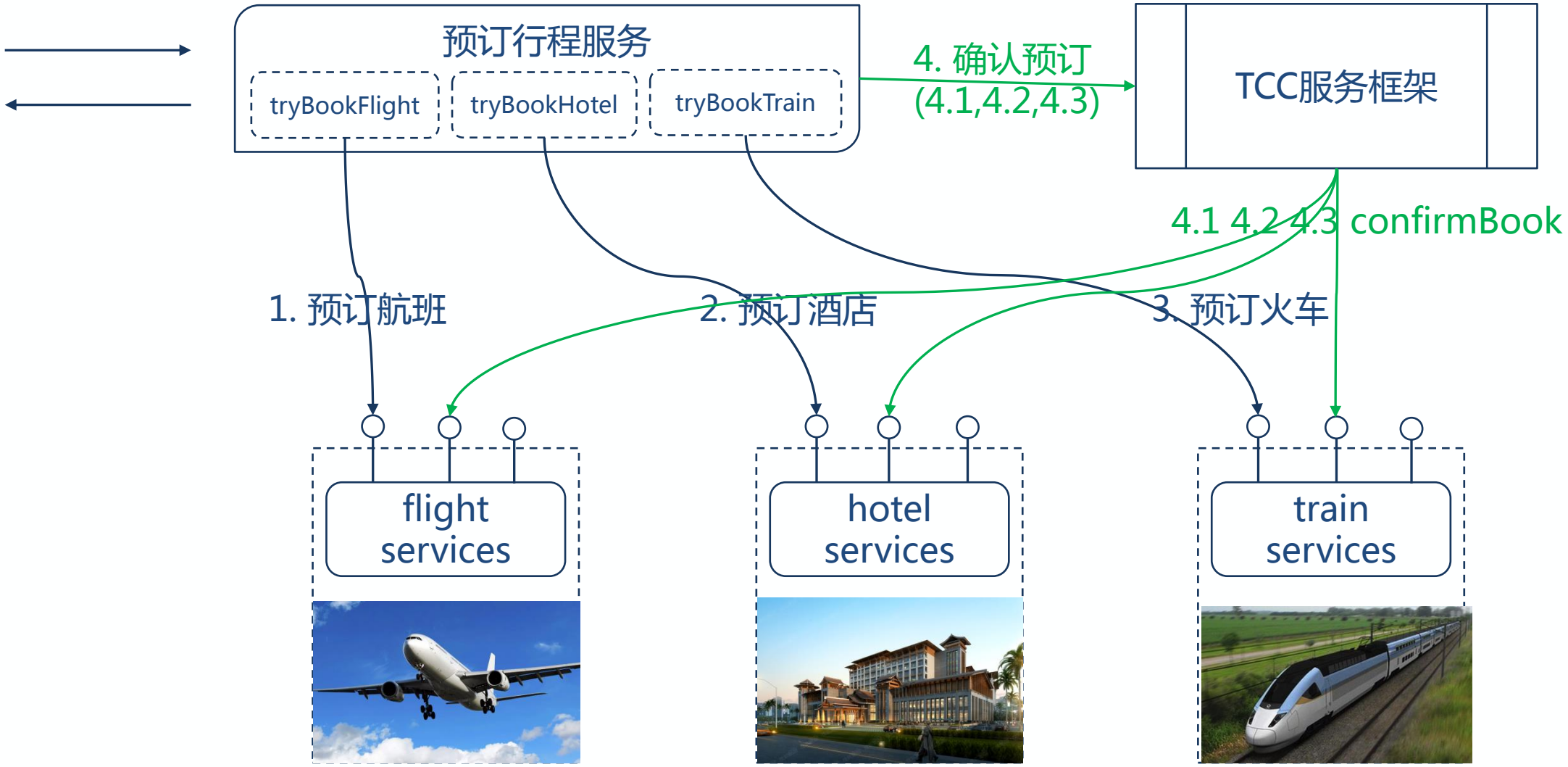
重试策略



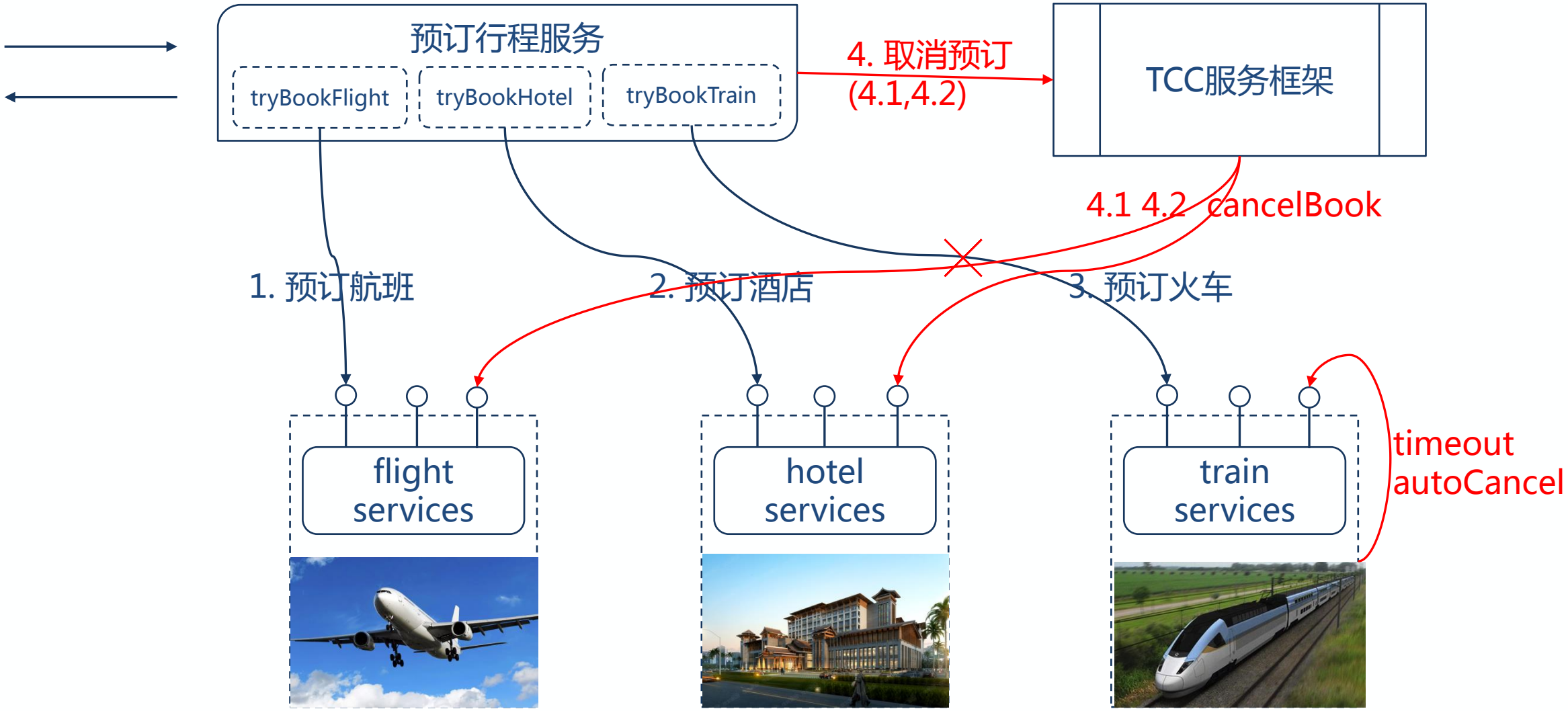
Try-Confirm-Cancel

TCC模式是优化的补偿模式

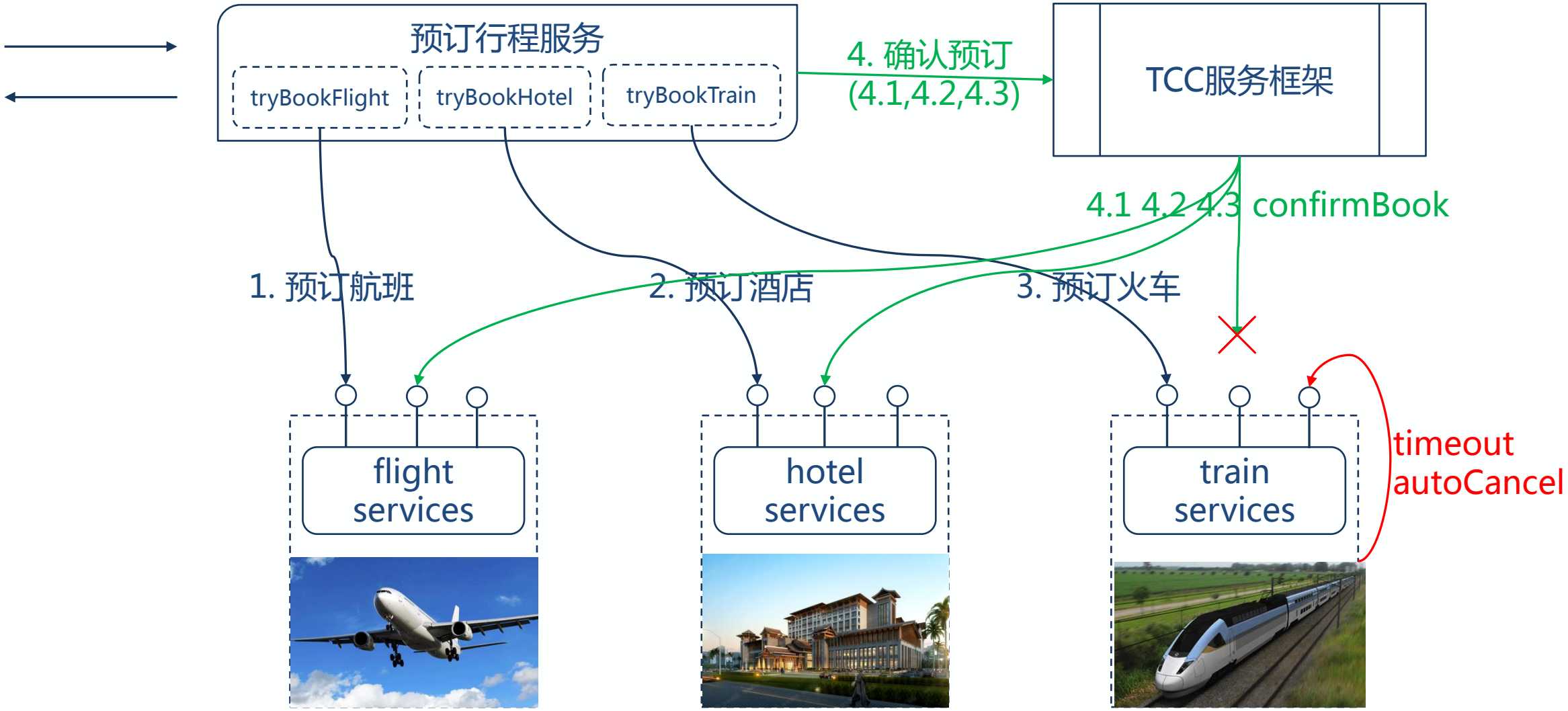
TCC模式-confirm



TCC模式-cancel



TCC模式-heuristic exception



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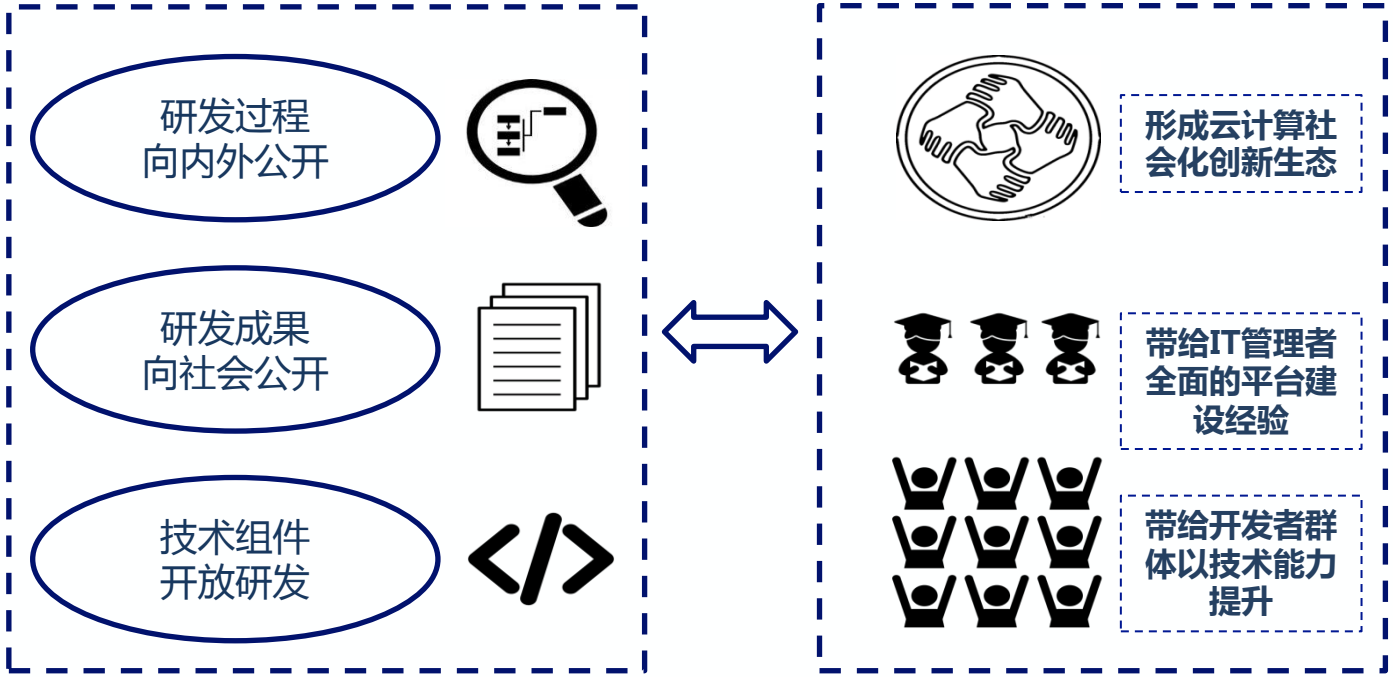
用开放，形成创新的社会生态



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用分享，降低创新的社会成本

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谢谢

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